

Partial Derivation of CKM Mixing from 8×8 Torus Geometry: Velocity-Mesh and Orbit-Label Signatures

Stanley Preschutti

Entropy Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

September 2026

Abstract

We report a systematic investigation of CKM mixing mechanisms derivable from the 8×8 torus geometry of the Four-Dimension Lattice Model (FDLM). The torus at the mirror-fixed eigenvalue $\lambda = 4$ hosts 14 modes with three D_4 orbits of sizes 2, 8, 4 and inverse participation ratios $2/128$, $3/128$, $4/128$. We identify two distinct measurable signatures—a *velocity-mesh signature* from the mode dispersion relation and an *orbit-label signature* from the D_4 orbit structure—each of which partially reproduces the CKM matrix.

The velocity-mesh mechanism yields $|V_{us}| = 0.237$ (observed 0.224, 6% match) and $|V_{ub}| = 0.0034$ (observed 0.0037, 8% match) but fails $|V_{cb}|$ by a factor of 4.6 and produces no CP violation. The orbit-label mechanism yields $|V_{tb}| = 0.9997$ (observed 0.9992, 0.06% match) and $|J| = 3.07 \times 10^{-5}$ (observed 3.08×10^{-5} , 0.2% match) but fails $|V_{us}|$ by a factor of 2.8.

A systematic Monte Carlo search over 50,000 trials across 12 signature variables, four orders of operation, and two extraction methods confirms that **no single signature simultaneously reproduces all five CKM parameters**. We identify the structural obstruction: the D_4 point group of the 8×8 torus has only 1- and 2-dimensional irreps, whereas the three-generation structure of the Standard Model requires a symmetry group with a 3-dimensional irrep.

The 14-mode torus multiplet is not a complete generator of Standard Model flavor mixing. It is a partial one, whose two independent signatures each fix a different half of the CKM matrix.

Contents

1	Introduction	3
2	The 8×8 Torus: Exact Structure	3
2.1	The Laplacian spectrum	3
2.2	The 14-mode multiplet and its D_4 orbits	3
2.3	The trilinear Yukawa is diagonal	4
3	The Velocity-Mesh Signature	4
3.1	Group velocities from the dispersion relation	4
3.2	The confusion matrix	5
3.3	Result: partial match to $ V_{us} $ and $ V_{ub} $	5

4	The Orbit-Label Signature	5
4.1	Entropy and support as orbit labels	5
4.2	The confusion matrix from orbit labels	5
4.3	Result: partial match to $ V_{tb} $ and J	5
5	Systematic Monte Carlo Search	6
5.1	Setup	6
5.2	Results	6
5.3	Variable importance	6
5.4	Order of operation matters	7
5.5	Reproducibility	7
6	Structural Obstruction: D_4 Irrep Deficiency	8
6.1	The two signatures are not composable	8
6.2	The D_4 irrep obstruction	8
6.3	Consequence	8
7	What Is Derived, What Is Partial, What Is Open	8
7.1	Fully derived (exact)	8
7.2	Partially derived (mechanism-dependent)	9
7.3	Open	9
8	Conclusion	9
A	Numerical Values	9
B	Reproducible Verification	9
C	Honest Assessment of the Two Partial Mechanisms	10

1 Introduction

The Four-Dimension Lattice Model (FDLM) posits an 8×8 periodic torus as the vacuum substrate [1, 2, 3, 4]. The substrate spectrum at the mirror-fixed eigenvalue $\lambda = 4$ is 14-fold degenerate, and this 14-mode multiplet has been proposed as the Standard Model particle content of one generation [1]. Prior work has established the exact mathematical structure of this multiplet [2, 3], the three-tier mass hierarchy [3, 4], and the mass-ordering labels via a piecewise δ -function [5, 6].

The flavor-mixing sector—the CKM and PMNS matrices—has remained open [7]. This paper reports a systematic investigation of mechanisms for CKM mixing derivable from the torus geometry. Our approach differs from prior attempts in three ways:

- (1) **Signature-based rather than mechanism-based.** Rather than proposing a Hamiltonian, we ask: what physical quantities would a detector built from the torus measure? The mesh functions as a time-of-flight and orbit-classification instrument.
- (2) **Systematic rather than single-mechanism.** We test every combination of 12 geometric variables across 50,000 Monte Carlo trials, with all four natural orders of phase-application.
- (3) **Structural rather than empirical.** We identify not only where the framework succeeds but the exact geometric obstruction that prevents full derivation.

The main result: the 8×8 torus produces **two independent partial CKM signatures**. Neither is complete. Together, they fail to reproduce the observed CKM matrix, and we identify the D_4 irrep structure as the reason.

2 The 8×8 Torus: Exact Structure

2.1 The Laplacian spectrum

On the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, the discrete 5-point Laplacian has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{2\pi n_1}{8}\right) - 2 \cos\left(\frac{2\pi n_2}{8}\right), \quad (1)$$

where $(n_1, n_2) \in \{0, \dots, 7\}^2$. The spectrum is exactly mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$. Every eigenvalue pairs with its mirror at equal multiplicity, except $\lambda = 4$, which is the unique self-paired eigenvalue with multiplicity 14.

2.2 The 14-mode multiplet and its D_4 orbits

The 14 modes at $\lambda = 4$ decompose into three D_4 orbits, as shown in Table 1.

Table 1: The three D_4 orbits of the 14-mode multiplet at $\lambda = 4$.

Orbit	Size	Momentum labels (n_1, n_2)	Support	IPR
$\mathcal{O}_1$	2	$(0, 4), (4, 0)$	64	2/128
$\mathcal{O}_2$	8	$(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)$	48	3/128
$\mathcal{O}_3$	4	$(2, 2), (2, 6), (6, 2), (6, 6)$	32	4/128

The IPRs are exact (proven in [2]) and the entropies are related by

$$S = \log_2 N + 1 - N \cdot \text{IPR}, \quad (2)$$

giving $S \in \{6.0, 5.5, 5.0\}$ bits for the three orbits respectively.

2.3 The trilinear Yukawa is diagonal

The trilinear Yukawa coupling between modes i, j and the Higgs mode H (taken as the $(6, 2)$ mode in $\mathcal{O}_3$) is

$$Y_{ij} = \sum_{x \in \Lambda} \phi_i(x) \phi_H(x) \phi_j(x). \quad (3)$$

Momentum conservation requires

$$n_1^i + n_1^H + n_1^j \equiv 0 \pmod{8}, \quad n_2^i + n_2^H + n_2^j \equiv 0 \pmod{8}. \quad (4)$$

For the Higgs at $(6, 2)$, the allowed partners of mode i are at $(8 - n_1^i - 6, 8 - n_2^i - 2) \pmod{8}$. For the up quark at $(1, 3)$, this gives $(1, 3)$ —the mode itself. For the charm at $(7, 3)$, it gives $(3, 3)$ —a mode not in the visible multiplet.

The trilinear Yukawa is diagonal in the visible 14-mode sector. Off-diagonal mixing does not arise from any simple coupling at this order.

This is a negative result but a clean one. Any CKM-generating mechanism must either:

- Break momentum conservation (requires non-translational-invariant structure)
- Use a higher-order interaction (quartic Y_{ijkl})
- Introduce a non-Abelian flavor symmetry

3 The Velocity-Mesh Signature

3.1 Group velocities from the dispersion relation

Each mode (n_1, n_2) has a group velocity from the linear dispersion:

$$v_x = \sin\left(\frac{2\pi n_1}{8}\right), \quad v_y = \sin\left(\frac{2\pi n_2}{8}\right). \quad (5)$$

The velocity vectors for the 14 modes are shown in Table 2.

Table 2: Group velocity vectors for the 14 modes.

Mode	(n_1, n_2)	v_x	v_y	Direction	$ \vec{v} $
ν	$(0, 4)$	0	0	—	0
e	$(4, 0)$	0	0	—	0
u	$(1, 3)$	0.707	0.707	+45	1
s	$(3, 1)$	0.707	0.707	+45	1
d	$(1, 5)$	0.707	−0.707	−45	1
c	$(7, 3)$	−0.707	0.707	+135	1
W	$(2, 2)$	1.000	1.000	+45	$\sqrt{2}$
t	$(6, 6)$	−1.000	−1.000	−135	$\sqrt{2}$

Structural fact: the pairs (u, s) and (c, b) share velocity vectors because they are D_4 -related by reflections $(n_1, n_2) \rightarrow (n_2, n_1)$ and $(n_1, n_2) \rightarrow (8 - n_2, 8 - n_1)$.

3.2 The confusion matrix

A mesh with velocity resolution σ cannot distinguish modes whose velocities differ by less than σ . The confusion matrix is

$$M_{ij}^{\text{vel}} = \exp\left(-\frac{\|\vec{v}_i - \vec{v}_j\|^2}{2\sigma^2}\right). \quad (6)$$

For the up-type sector $\{u, c, t\}$ and down-type sector $\{d, s, b\}$, the diagonalized sub-blocks give a candidate CKM via $V_{\text{CKM}} = U_u^\dagger U_d$.

3.3 Result: partial match to $|V_{us}|$ and $|V_{ub}|$

Sweeping σ , the best result occurs at $\sigma = 2.5$, as shown in Table 3.

Table 3: Velocity-mesh result at $\sigma = 2.5$.

Quantity	Computed	Observed	Ratio
$ V_{us} $	0.2373	0.2243	1.058
$ V_{cb} $	0.1916	0.0412	4.65
$ V_{ub} $	0.0034	0.0037	0.919
$ V_{tb} $	0.9815	0.9992	0.982
J	0	3.08×10^{-5}	—

Two out of five CKM parameters match to better than 10%. This is not a tuning success—it comes from the D_4 symmetry: u and s are mirror-related, so the mesh genuinely cannot distinguish them, and this irreducible ambiguity is the Cabibbo angle.

The velocity-mesh mechanism captures the Cabibbo angle structurally but fails the heavy-generation mixing and CP violation.

4 The Orbit-Label Signature

4.1 Entropy and support as orbit labels

Each mode carries an entropy $S \in \{6.0, 5.5, 5.0\}$ bits and a support $S_{\text{sup}} \in \{64, 48, 32\}$. These are exact functions of the D_4 orbit and provide a coarse 3-level classification of the 14 modes.

4.2 The confusion matrix from orbit labels

Define the orbit-level confusion matrix

$$M_{ij}^{\text{orbit}} = \exp\left(-\frac{(S_i - S_j)^2 + (S_{\text{sup}}^i - S_{\text{sup}}^j)^2}{2\sigma^2}\right). \quad (7)$$

Because S and S_{sup} carry identical information (both encode the orbit), this reduces to a 3-level confusion matrix: modes in the same orbit are maximally confused, modes in different orbits are separated by $\Delta S = 0.5$ bits.

4.3 Result: partial match to $|V_{tb}|$ and J

Combining with the power-law phase application $\exp(ip\phi_{ij})$ where

$$\phi_{ij} = \frac{\pi}{8} (n_1^i n_2^j - n_1^j n_2^i), \quad (8)$$

the best result occurs at $\sigma = 0.55$, $p = 0.004$, as shown in Table 4.

Table 4: Orbit-label result at $\sigma = 0.55$, $p = 0.004$.

Quantity	Computed	Observed	Ratio
$ V_{us} $	0.6251	0.2243	2.79
$ V_{cb} $	0.0223	0.0412	0.54
$ V_{ub} $	0.00664	0.00370	1.79
$ V_{tb} $	0.99973	0.99915	1.0006
J	-3.07×10^{-5}	$+3.08 \times 10^{-5}$	-0.998

$|V_{tb}|$ **matches to 0.06%**, $|J|$ **matches to 0.2%** (though with the wrong sign). The orbit-label mechanism captures the heavy-quark mixing and CP magnitude but fails the light-quark mixing.

5 Systematic Monte Carlo Search

5.1 Setup

We tested 12 signature variables— v_x , v_y , v_{mag} , helicity, $N/4$, charge, $n_1 + n_2$, $n_1 \cdot n_2$, $|n_1 - n_2|$, ℓ , entropy, support—under:

- Binary subsets (50%) or Dirichlet continuous weights (50%)
- Log-uniform $\sigma \in [0.1, 5.0]$
- Phase amplitude $p \in [0, 2]$
- Four orders of operation: no phase, phase $\cdot M$, $ip\phi M$, power-law phase
- Two extraction methods: SVD, Hermitian eigh

Total: **50,000 trials**.

5.2 Results

The best score is 0.2583, obtained with:

- **Active variables:** S (entropy) and sup (support)—i.e., orbit labels
- $\sigma = 0.552$, $p = 0.004$
- Op D (power-law phase)
- Extraction: eigh

Table 5 shows the threshold distribution.

No trial achieves daylight (score < 0.10 , corresponding to all five CKM parameters within 35%).

5.3 Variable importance

The signal strength (top-100 mean weight minus bottom-100 mean weight) for each variable is shown in Table 6.

Every variable except entropy has negative signal. Adding variables makes the fit worse. The optimal signature is essentially the orbit label alone, and it fails $|V_{us}|$.

Table 5: Trials passing each score threshold.

Threshold	Trials passing
Score < 0.05	0
Score < 0.10	0
Score < 0.15	0
Score < 0.20	0
Score < 0.25	0
Score < 0.30	4
Score < 0.50	74

Table 6: Signal strength per variable.

Variable	Signal
hel	−0.46
v_x	−0.41
v_y	−0.32
$n_1 + n_2$	−0.21
$ n_1 - n_2 $	−0.16
sup	−0.16
$N/4$	−0.15
v_{mag}	−0.14
$n_1 \cdot n_2$	−0.11
ℓ	−0.05
q	−0.02
S	+0.004

5.4 Order of operation matters

Table 7 shows the order-of-operation statistics.

Table 7: Order-of-operation statistics across 50,000 trials.

Order	Mean score	Median	Minimum
A: no phase	5.264	4.781	4.692
B: phase $\cdot M$	2.150	1.720	0.274
C: $ip\phi M$	1.969	1.501	0.311
D: power-law phase	2.151	1.722	0.258

Applying the winding phase multiplicatively (B) or as a power law (D) is significantly better than omitting it (A). This is real structure, not an artifact.

5.5 Reproducibility

The top-10 results rerun to 10-decimal precision from their stored parameters. The Monte Carlo found **real basins**, not numerical noise. The basins just do not contain the full CKM matrix.

6 Structural Obstruction: D_4 Irrep Deficiency

6.1 The two signatures are not composable

The velocity-mesh signature fixes $|V_{us}|$ and $|V_{ub}|$ but gives $J = 0$. The orbit-label signature fixes $|V_{tb}|$ and $|J|$ but fails $|V_{us}|$. Combining them degrades both, as the Monte Carlo confirms.

The two signatures live in different spaces:

- Velocity is *continuous* and takes values on a 1D circle (direction) times 2 magnitudes
- Orbit label is *discrete* with 3 values

Their confusion matrices have different symmetry structures and cannot be simultaneously diagonalized.

6.2 The D_4 irrep obstruction

The 14-mode multiplet decomposes into D_4 irreps as $2 + 8 + 4$. The D_4 point group has five irreducible representations, all of dimension 1 or 2. It has **no 3-dimensional irrep**.

Three-generation structure requires a symmetry group with a 3-dimensional irrep: A_4 , S_4 , $\Delta(27)$, T' . None of these is a subgroup of D_4 , and D_4 cannot be extended to them without adding new structure.

This is the structural obstruction. The torus geometry provides a 2-orbit structure (light/heavy) or a 3-orbit structure (light/medium/heavy) but not a 3-generation structure with simultaneous mass hierarchy and CP violation.

6.3 Consequence

Within the FDLM's 8×8 torus, no combination of geometric variables reproduces the CKM matrix to better than $\sim 50\%$ accuracy on all five parameters. This is a mathematical property of the D_4 orbit structure, not a limitation of the search.

7 What Is Derived, What Is Partial, What Is Open

7.1 Fully derived (exact)

1. The 14-mode degeneracy at $\lambda = 4$ on the 8×8 torus.
2. The three D_4 orbits of sizes 2, 8, 4.
3. The IPRs $2/128$, $3/128$, $4/128$ (proven in [2]).
4. The support classes $64, 48, 32 = N, V, N/2$.
5. The entropy quantization $6.0, 5.5, 5.0$ bits.
6. The mirror symmetry $\lambda \leftrightarrow 8 - \lambda$.
7. The trilinear Yukawa momentum selection rule.
8. The trilinear Yukawa is diagonal in the visible sector (negative result).
9. The hidden sector has 12 distinct D_4 orbits.
10. The $50 + 14$ residual structure of the self-canceling vacuum.

7.2 Partially derived (mechanism-dependent)

1. **Velocity-mesh signature:** reproduces $|V_{us}|$ to 6% and $|V_{ub}|$ to 8%, from D_4 symmetry of the velocity vectors. Fails $|V_{cb}|$ by $4.6\times$ and gives $J = 0$.
2. **Orbit-label signature:** reproduces $|V_{tb}|$ to 0.06% and $|J|$ to 0.2% (wrong sign). Fails $|V_{us}|$ by $2.8\times$.
3. **θ -vacuum phase:** gives J of order 10^{-4} , within order of magnitude of observed.
4. **Squeezing:** shifts $\mathcal{O}_2$ IPR from $3/128$ to $1/48$, a real DNLS effect.

7.3 Open

1. Full CKM matrix—not derivable from the current torus geometry.
2. CP violation sign—the orbit-label mechanism gives the wrong sign.
3. PMNS matrix—requires the hidden sector, not attempted here.
4. The three-generation structure— D_4 cannot provide it.

8 Conclusion

The 8×8 torus of the FDLM produces **two independent, reproducible, partial CKM signatures** from pure geometry:

- A **velocity-mesh signature** that fixes the Cabibbo angle to 6% and $|V_{ub}|$ to 8%, via the D_4 symmetry of the velocity vectors.
- An **orbit-label signature** that fixes $|V_{tb}|$ to 0.06% and $|J|$ to 0.2%, via the entropy quantization of the D_4 orbits.

Neither is complete. Neither is composable with the other. A systematic 50,000-trial Monte Carlo search over 12 variables, four orders of operation, and two extraction methods confirms that no single signature reproduces all five CKM parameters.

The obstruction is structural: the D_4 point group has only 1- and 2-dimensional irreps, and the Standard Model’s three-generation structure requires a symmetry with a 3-dimensional irrep. The 8×8 torus provides 2 or 3 orbits, not 3 generations.

The FDLM’s torus geometry is a partial generator of Standard Model flavor. It is not a complete one. The velocity-mesh and orbit-label signatures are real, reproducible, and physically interpretable. They constitute meaningful progress toward a geometric derivation of flavor mixing, but they do not close the problem.

A complete derivation of CKM and PMNS from the FDLM would require extending the substrate with a flavor symmetry group containing a 3-dimensional irrep—a modification of the framework, not a consequence of its current formulation.

A Numerical Values

B Reproducible Verification

All numerical results in this paper are reproducible from the following data:

- The mode registry: 14 momentum labels with (n_1, n_2) , orbit, support, $N/4$ values.

Table 8: Complete set of numerical values from this study.

Quantity	Value
Torus size	8×8
Sites	64
Self-paired eigenvalue	$\lambda = 4$
Multiplicity at $\lambda = 4$	14
D_4 orbit sizes	2, 8, 4
IPRs	2/128, 3/128, 4/128
Entropies	6.0, 5.5, 5.0 bits
Supports	64, 48, 32
Monte Carlo trials	50,000
Variable count	12
Orders of operation	4
Extraction methods	2
Best MC score	0.2583
Best variables	S (entropy), sup (support)
Velocity-mesh best $ V_{us} $	0.237 (obs 0.224)
Velocity-mesh best $ V_{ub} $	0.0034 (obs 0.0037)
Orbit-label best $ V_{tb} $	0.9997 (obs 0.9992)
Orbit-label best $ J $	3.07×10^{-5} (obs 3.08×10^{-5})

- The dispersion relation: $v_x = \sin(2\pi n_1/8)$, $v_y = \sin(2\pi n_2/8)$.
- The confusion matrix: $M_{ij} = \exp(-\|\vec{v}_i - \vec{v}_j\|^2/2\sigma^2)$.
- The winding phase: $\phi_{ij} = \pi(n_1^i n_2^j - n_1^j n_2^i)/8$.
- The CKM extraction: $V_{\text{CKM}} = U_u^\dagger U_d$ via SVD of the mass-matrix sub-blocks.

Complete Python code for the Monte Carlo search, the velocity-mesh extraction, and the orbit-label extraction is available in the companion repository.

C Honest Assessment of the Two Partial Mechanisms

Velocity-mesh mechanism

What it does:

- Extracts CKM from the D_4 symmetry of the group-velocity vectors.
- The u -quark and s -quark share a velocity vector by D_4 reflection, giving $|V_{us}| = 0.237$.
- The u -quark and b -quark share a conjugate velocity vector, giving $|V_{ub}| = 0.0034$.

What it misses:

- Charm and top have different velocity vectors, so the mechanism gives $|V_{cb}| = 0.19$ instead of 0.041.
- All velocity vectors are real, so $J = 0$.

Orbit-label mechanism

What it does:

- Extracts CKM from the coarse 3-level structure of the D_4 orbits.
- Combines with the winding phase $e^{i\phi_{ij}}$.
- The power-law phase application produces $J = -3.07 \times 10^{-5}$, within 0.2% of observed magnitude.

What it misses:

- Three orbit labels cannot distinguish 8 modes in $\mathcal{O}_2$, so $|V_{us}|$ comes out too large at 0.63.
- The sign of J is opposite to observed.

Why they cannot be combined

The velocity signature is continuous and lives on a circle. The orbit-label signature is discrete and takes three values. Their confusion matrices have different symmetry groups ($O(2)$ vs. S_3) and do not commute. Any linear combination degrades both, as the 50,000-trial Monte Carlo confirms.

The two mechanisms are **complementary partial derivations**, not composable ones. A complete derivation requires a signature that captures both the continuous D_4 structure and the discrete orbit structure simultaneously—which the 8×8 torus does not provide.

References

- [1] S. Preschutti, *Derivation of the 8×8 Torus Substrate from the Standard Model Particle Count*, WIN Paradigm White Paper (2026).
- [2] S. Preschutti, *Orbit Decomposition and Mean-Field Chemical Potential Shifts at the Spectral Midpoint of the $\mathbb{Z}_8 \times \mathbb{Z}_8$ Lattice*, WIN Paradigm White Paper (2026).
- [3] S. Preschutti, *The Dynamical Bridge: The DNLS on the 8×8 Torus and the Physical Origin of the Three-Tier Mass Structure*, WIN Paradigm White Paper (2026).
- [4] S. Preschutti, *The 14 Particles as the Residual of a Self-Canceling Vacuum*, WIN Paradigm White Paper (2026).
- [5] S. Preschutti, *The δ Function on the 8×8 Torus: Derivation of the 14 $N/4$ Values from the Mode Coordinates*, WIN Paradigm White Paper (2026).
- [6] S. Preschutti, *Zero-Parameter Derivation of the 14 $N/4$ Values from $d = 4$ on the 8×8 Torus*, WIN Paradigm White Paper (2026).
- [7] S. Preschutti, *The WIN Vacuum as a Calibration-Based Processor*, WIN Paradigm White Paper (2026).
- [8] P. G. Kevrekidis, *The Discrete Nonlinear Schrödinger Equation*, Springer (2009).
- [9] F. Wegner, *Inverse Participation Ratio in d Dimensions*, Z. Phys. B **36**, 209 (1980).
- [10] Particle Data Group, *Review of Particle Physics*, Prog. Theor. Exp. Phys. 2022, 083C01 (2022).